

Agent-

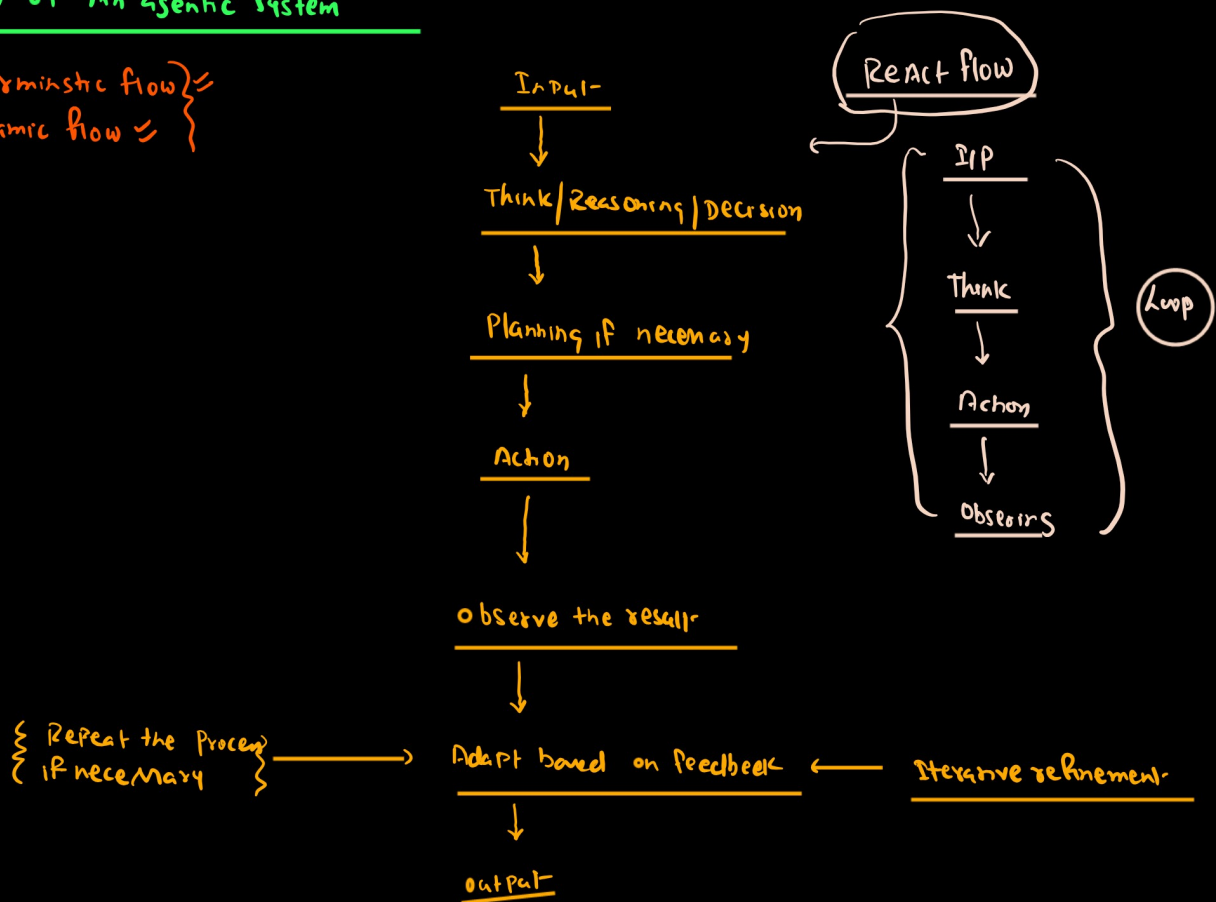
- ① What is an AI Assistant ✓
- ② " RAG Based assistant ✓
- ③ What is an Agent? ✓
- ④ What is Agentic AI? ✓
- ⑤ What is multiAgent ✓
- ⑥ What is Deep Agentic System ✓
- ⑦ When & Why you should use an AI Agent ✓

As an AI eng.

Shallow Agent | Deep Agent

Flow of an Agentic System

{ Deterministic flow }
{ Dynamic flow }



Can be implemented

- | | |
|--|---|
| ① <u>Pure Python</u> → <u>from scratch</u> | ⑥ <u>CrewAI</u> ^{3rd} → |
| <u>Winner</u> → ② <u>Langgraph</u> → <u>Langchain</u> ^{1st} | ⑦ <u>SmolAgent</u> → <u>HuggingFace</u> |
| → ③ <u>OpenAI Agent SDK</u> → <u>OpenAI</u> ^{2nd} | ⑧ <u>N8N</u> → <u>Node framework</u> |
| → ④ <u>Google ADK</u> → <u>Google</u> ^{2nd} | |
| ⑤ <u>Autogen</u> → <u>microsoft</u> ^{3rd} | |

- ① Planning | Dynamic Planning
- ② Execution
- ③ Task decomposition
- ④ Reflection
- ⑤ self correction | improvement | healing | critic
- ⑥ Reasoning
- ⑦ Parallelization
- ⑧ Orchestration
- ⑨ human in loop
- ⑩ feedback aware optimization

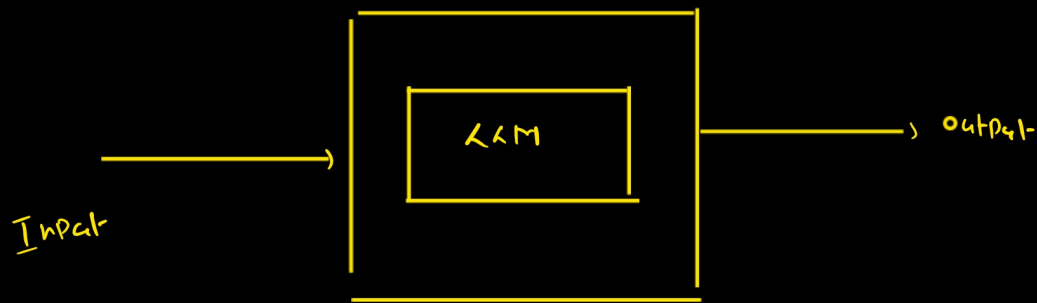
It is Related to the Agent's Flow

{ maximum iteration limit
Time out
stop condition
Token / cost budget }

- ① Pydantic | enum | typehint
- ② Json Schema
- ③ Async Python
- ④ Structure OIP
- ⑤ tool engineering
- ⑥ memory management
- ⑦ statemanagment

- ⑧ Prompting
- ⑨ Context engineering
- ⑩ mcp | A2A Protocol
- ⑪ evaluation | testing of Agent -
- ⑫ observability
- ⑬ Guardrails | Security
- ⑭ caching

1 AI Assistant → Simple AI Assistant

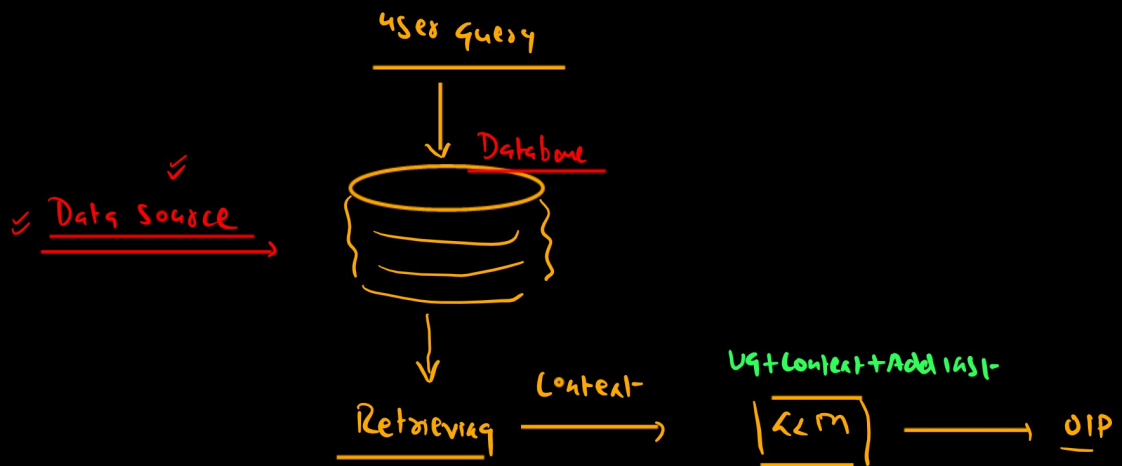


Generate response → model's Available Knowledge

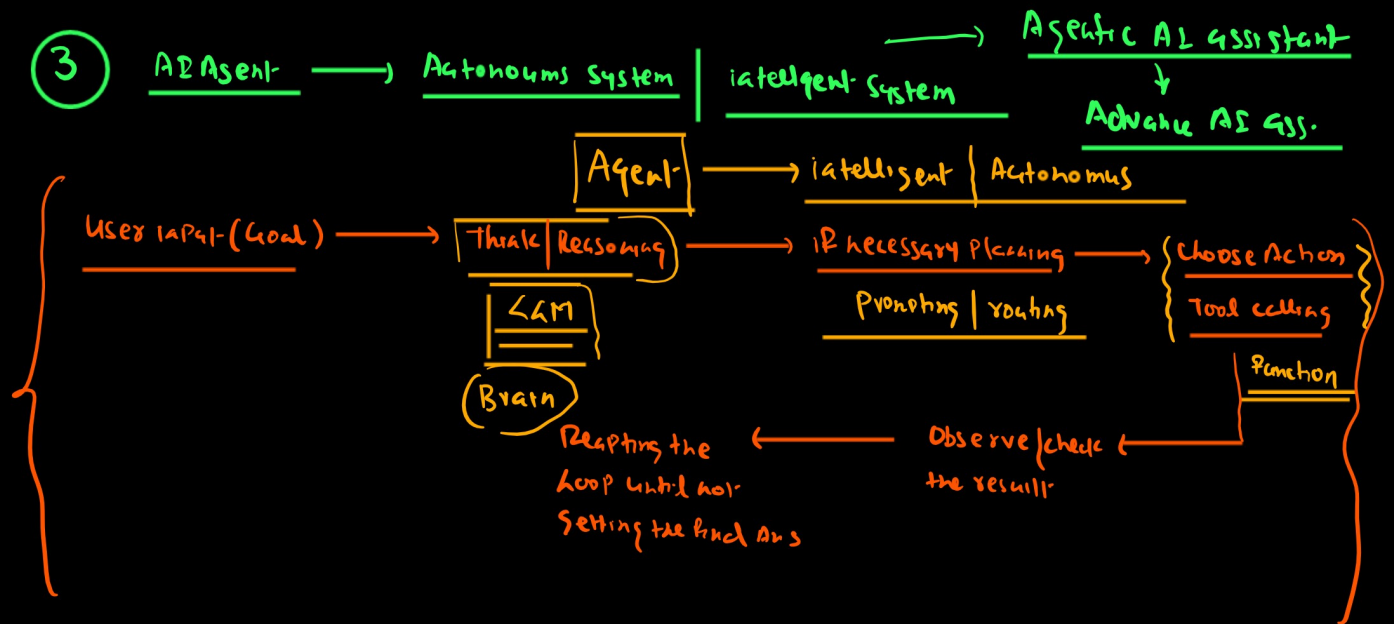
Train

{
 ex:- What is Tesla?
 Ans: Tesla is an EV company
}

2 RAG Assistant → RAG enable AI Assistant



A RAG Assistant is an AI assistant that retrieves relevant external knowledge and uses that information to generate a more grounded response.



Example

User says: "Analyze Tesla and prepare an investment research report."

I need current financial information.



Use Web Search

I need historical stock data.



Use Finance API

I need calculations.



Use Python

I need Tesla's annual report.



Search Documents

Do I have enough information?



No → Search again

Yes



Generate Report

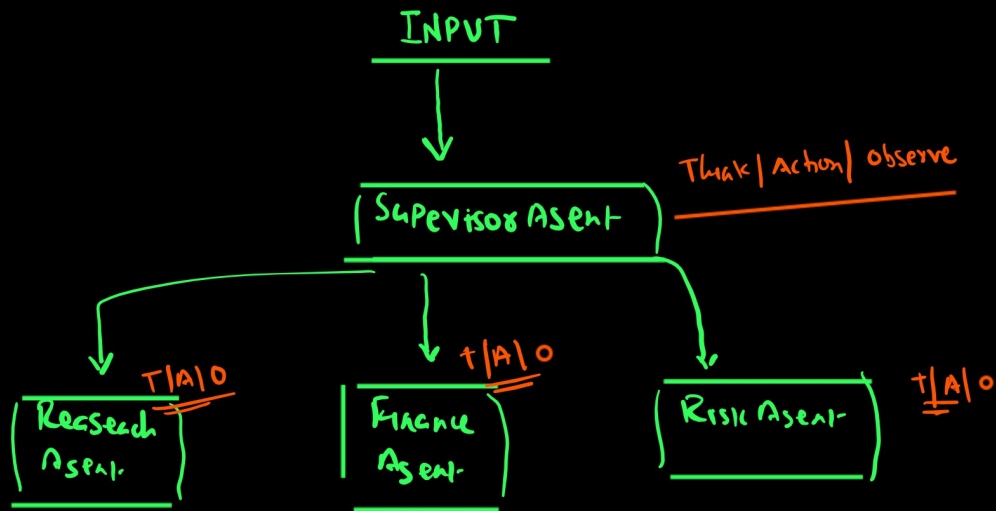
An AI Agent is an AI system that can autonomously decide and execute actions using tools and environmental feedback to achieve a goal.

4

Multi-Agent System

A Multi-Agent System uses multiple agents instead of one agent.

Each agent can specialize in a particular job.



Tesla example

User: "Analyze Tesla and prepare a detailed investment research report."

A supervisor may delegate:

Research Agent Research:

Tesla products

Market

Competitors

Recent developments

Financial Agent Analyze:

Revenue

Profit

Cash flow

Margins

Risk Agent Analyze:

Regulatory risks

Competition

Supply-chain risks

Market Agent Analyze:

EV market

Competitors

Market share

Writer Agent

Combine everything into the report. So instead of:

One Agent

↓

Does everything

we have:

Multiple Specialized Agents

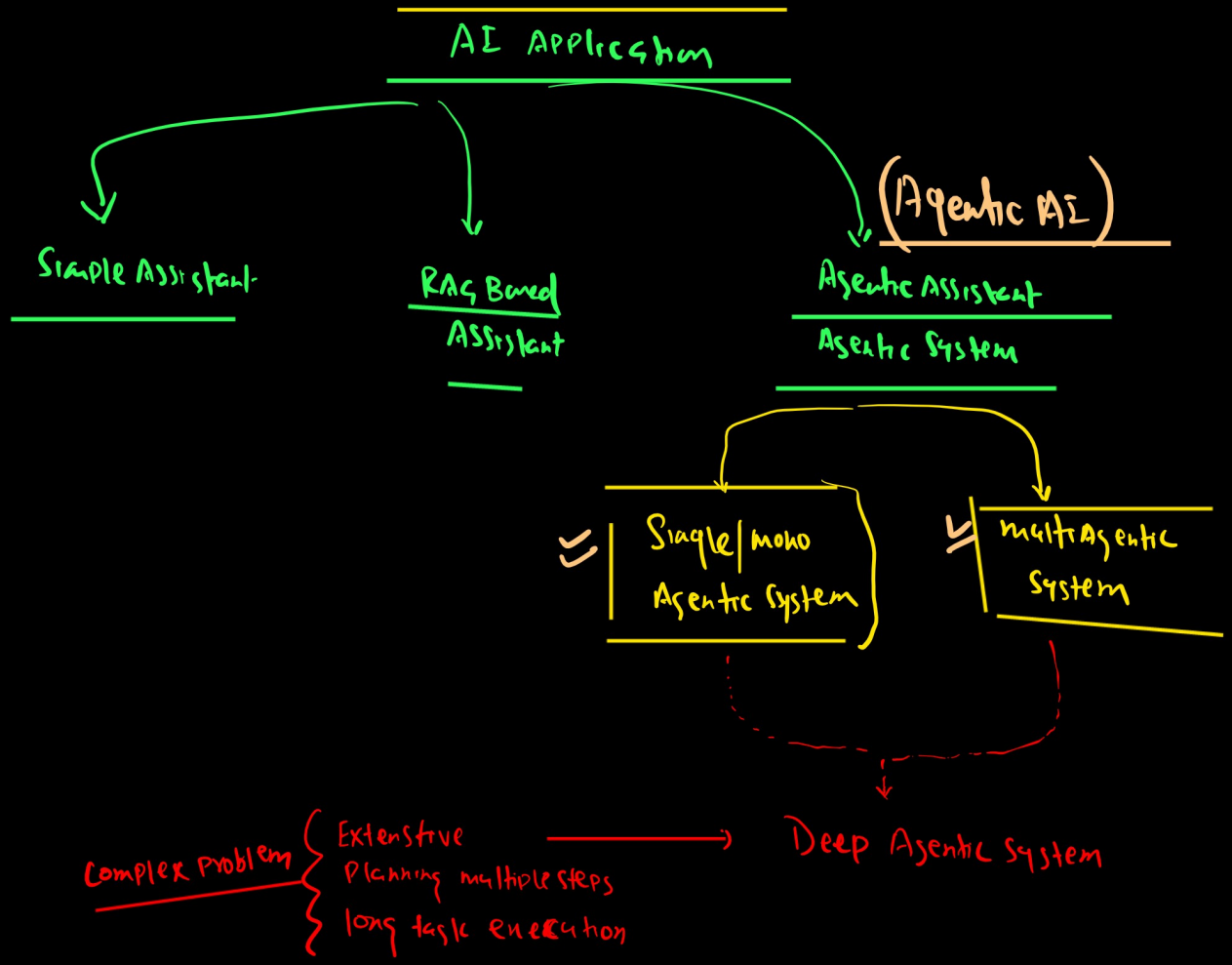
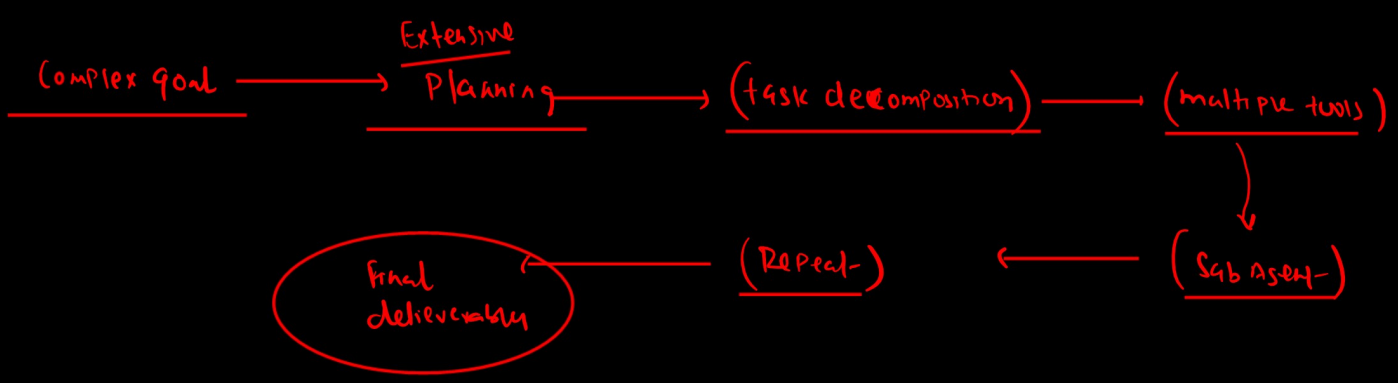
↓

Collaborate / Delegate

↓

Complete Goal

5 (Deep Agent) → (Large & Long running)
(Complex Problem)



Tesla example

A Deep Agent could first generate:

TODO

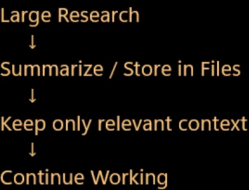
- 1. Download Tesla reports
- 2. Extract financial data
- 3. Research competitors
- 4. Analyze EV market
- 5. Calculate financial ratios
- 6. Generate charts
- 7. Research risks
- 8. Compare companies
- 9. Verify sources
- 10. Write report

Then work through those tasks.

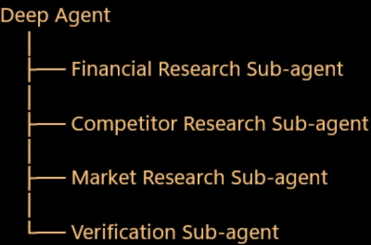
During execution it might create:

/workspace
/tesla_financials.csv
/competitor_analysis.md
/market_research.md
/risk_analysis.md
/charts/
/sources.md
/final_report.docx

And if the context becomes too large:



It could also delegate:



Langgraph



1

Introduction of Langgraph

Langgraph vs Langchain vs Langsmith

One Program

→ Langgraph vs normal Python code



2

Langgraph Architecture } tools

3

StateManaged-

4

Stategraph API vs Graph API vs Functional API

5

Send_API () | map-reduce

6

Command_APP ()

7

Persistence / Checkpoint

8

Memory

9

Launch loop

10

Streaming

11

Multiagent

12

Caching

13

Fault tolerance

14

Async Langgraph

Graph visualization

Subgraph